

什么是本体？

Fabian Neuhaus

摘要

在知识工程领域，“本体”通常按照 Gruber 的传统定义为一个“概念化的显式规范”。这个定义存在多种变体。在本文中，我们论证（除一个显著例外外）这些定义没有解释价值，因为它们违反了良好定义的基本规则之一：定义陈述（定义项）应该比被定义的术语（被定义项）更清晰。在本文中，我们提出了“本体”的不同定义，并讨论了它如何帮助解释各种现象：本体改变的能力、词汇选择的作用、注释的意义、协作本体开发的可能性，以及本体概念主义与本体实在论之间的关系。

1 引言

1992 年，Tom Gruber 提出了如下定义：“本体是一个概念化的规范” [4]。存在几种变体，通常添加形容词进一步描述规范（例如，“形式化”、“显式”）或概念化（例如，“共享”）（参见第 5 节中相关工作的讨论）。这些定义没有帮助，因为它们违反了良好定义的基本规则之一：定义陈述（定义项）应该比被定义的术语（被定义项）更清晰。只要“概念化”比“本体”更模糊，任何试图将“本体”定义为一类“概念化的规范”的尝试都是一种智力安慰剂：它让我们感觉它提供了对本体性质的更好把握，但没有智力进步，因为它缺乏解释价值（有关详细信息，请参见第 2 节）。

鉴于定义“本体”存在困难，人们可能会得出结论：一个恰当的定义并非真正必要。这似乎是 [7]，中 Gruber 的观点，他在讨论对他原始定义的反意见后写道：

从更务实的角度来看，可以说本体是工程学的一个工具和产物，并因此由其用途来定义。

What is an Ontology?

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Abstract

In the knowledge engineering community “ontology” is usually defined in the tradition of Gruber as an “explicit specification of a conceptualization”. Several variations of this definition exist. In the paper we argue that (with one notable exception) these definitions are of no explanatory value, because they violate one of the basic rules for good definitions: The defining statement (the definiens) should be clearer than the term that is defined (the definiendum). In the paper we propose a different definition of “ontology” and discuss how it helps to explain various phenomena: the ability of ontologies to change, the role of the choice of vocabulary, the significance of annotations, the possibility of collaborative ontology development, and the relationship between ontological conceptualism and ontological realism.

1 Introduction

In 1992 Tom Gruber proposed the following definition “An ontology is a specification of a conceptualization” [4]. Several variants exist that usually add adjectives further describing the specification (e.g., “formal”, “explicit”) or the conceptualization (e.g., “shared”) (see discussion of related work in Section 5). These definitions are not helpful because they violate one of the basic rules for good definitions: the defining statement (the definiens) should be clearer than the term that is defined (the definiendum). As long as “conceptualization” is murkier than “ontology”, any attempt of defining “ontology” as a kind of “specification of a conceptualization” is an intellectual placebo: it makes us feel like it provides a better grasp of the nature of ontologies, but there is no intellectual progress, because it lacks explanatory value (see Section 2 for details).

Given the difficulties in defining “ontology” one may come to the conclusion that a proper definition is not really needed. This seems to be the sentiment of Gruber in [7], who writes after discussing an objection to his original definition:

Taking a more pragmatic view, one can say that ontology is a tool and product of engineering and thereby defined by its use.

然而，对于应用本体学家来说，轻易放弃这一点会相当尴尬。毕竟，我们本应是定义术语的专家。因此，我们应该能够为我们自己的研究对象提供一个令人满意的定义。本文的目的是提出一个关于“本体”的新定义，可以非正式地表述为：

非正式定义 1 一个特定领域的本体是一个提供文档

- 1. 用于描述特定领域的词汇表，
- 2. 文档化词汇表的注释，和
- 3. 用于词汇的逻辑理论（由公理和定义组成），以这种方式，这三个元素共同使本体的合格用户能够确定其预期解释。

在详细介绍并将在第 4 节中形式化此提案之前，我们首先讨论一些现象，这些现象应该通过“本体”的定义来阐明（见第 3 节）。我们关于本体的解释的核心是注释的作用，注释不仅为词汇的“含义”提供了基础，而且通常提供比逻辑理论所捕获的更多信息。我们解释的一个主要优势在于，它解释了不同语言的本体版本之间的关系（例如，BFO 在 FOL 和 OWL 中）。它进一步使我们能够清晰地分析本体中的某些错误，例如注释与本体的逻辑理论之间的不匹配。所提出的定义具有额外的优势，即它与概念主义和本体现实阅读都兼容。实际上，可以证明，给定一个领域的正确概念化，任何正确概念化的正确本体都是该领域的正确本体（第 4.5 节）。

2 荒芜与肥沃的定义

我们可以自由地以任何方式定义术语，但有些定义比其他定义更好。为了说明这一点，让我们考虑“英寸”的两个不同定义。

示例 1

E.1a 英寸是三粒大麦的长度。

E.1b 英寸是光在真空中传播 $\frac{1}{11802852677}$ 秒所经过的路径长度。

示例 E1.a 基于英国爱德华二世在14世纪颁布的一项法规。示例 E1.b 大致符合当代定义，即一英寸等于2.54厘米，而米是根据光速定义的。这些定义中哪个更好？答案似乎很明显，因为 E1.b 提供了更好的测量分辨率，而 E1.a 不允许进行测量

However, it would be rather embarrassing for applied ontologists to give up that easily. After all, we are supposed to be the experts on defining terms. Thus, we should be able to provide a satisfactory definition of our own subject. The purpose of this paper is to propose a novel definition of “ontology” which may be stated informally as:

Informal Definition 1 *An ontology of a given domain of interest is a document that provides*

- 1. *a vocabulary for describing the domain of interest,*
- 2. *annotations that documents the vocabulary, and*
- 3. *a logical theory (consisting of axioms and definitions) for the vocabulary, in a way that these three elements together enable a competent user of the ontology to ascertain its intended interpretation.*

Before this proposal is presented in detail and is formalized in Section 4, we first discuss some of the phenomena, which should be illuminated by a definition of “ontology” (see Section 3). Central to our account of ontologies is the role of annotations, which often not only grounds the ‘meaning’ of the vocabulary, but frequently provides much more information than is captured by the logical theory. One major benefit of our account is that it explains the relationship between versions of an ontology in different languages (e.g., BFO in FOL and OWL). It further enables us to clearly analyze certain errors in ontologies, e.g., a mismatch between the annotations and the logical theory of an ontology. The proposed definition has the additional benefit of being compatible with both conceptualism and a realistic reading of ontologies. Indeed, one can prove that given a correct conceptualization of a domain, any correct ontology of the conceptualization, is a correct ontology of the domain (Section 4.5).

2 Barren and Fertile Definitions

We are free to define terms in any way we want, but some definitions are better than others. To illustrate the point lets consider two different definitions of “inch”.

Example 1

E.1a An inch is the length of three barley corns.

E.1b An inch is the length of the path travelled by light in a vacuum within $\frac{1}{11802852677}$ seconds.

Example E1.a is based on a statute by Edward II of England in the 14th century. Example E1.b approximates the contemporary definition, according to which an inch is 2.54 cm and a meter is defined in terms of the speed of light. Which of these definitions is better? The answer seems obvious, because E1.b provides a better measurement resolution, since E1.a does not allow measuring

小于麦粒大小的长度差异。此外，E1.b 能够实现更高的测量精度，因为它避免了由于麦粒尺寸变化而引起的测量不确定性。

然而，设想一位有远见的自然哲学家曾向爱德华二世提议使用E1.b 而不是 E1.a。由于亚里士多德讨论过真空的存在以及光速的问题¹，这些概念在当时学者中应该是熟悉的。例如，在13世纪，维特罗曾考虑过光在真空中传播可能比在更稠密的介质中传播得更快 [14]。因此，假设有位阿拉伯数学家帮助推广了阿拉伯数字和分数，E1.b 就不会超出爱德华二世同时代人的概念范围。但爱德华二世是否应该选择 E1.b 而不是 E1.a 呢？

在一个农业中世纪社会里，没有人拥有精确的时钟，更不用说技术来测定光在某一秒的几分之一内传播多远。因此，爱德华二世同时代的人不可能应用 E1.b 并用它来确定物体的长度。所以，如果爱德华二世接受了 E1.b，这个决定对“英寸”的实际使用不会产生任何影响。由于英寸已经确立为一种计量单位，人们会继续按照他们一直理解的方式使用“英寸”，同时口头上承认这个定义。因此，皇家定义不会促进长度测量的标准化。相比之下，E1.a 提供了一个“英寸”的定义，这个定义对农业社会的人们来说容易理解和使用。

这个例子说明，定义的肥沃性取决于其使用的语境。肥沃性指的是人们接受一个定义后能够利用它的程度，例如，用来对某些现象进行分类或解决问题，或进行其他具体的智力过程。在21世纪的语境下，E1.b比E1.a更肥沃，因为它能更好地测量长度，并将其系统地与国际单位制中的其他测量单位联系起来。然而，在一个中世纪社会的语境下，E1.b将毫无用处；它将是无用的。这是因为在爱德华国王同时代的人已经比E1.b的定义项对“英寸”（被定义项）有更好的理解的情况下，这种情况才会发生。

大多数“本体”的定义与E1.b对爱德华二世国王来说一样贫瘠，原因也相同。对于该领域的初学者来说，通过举例和应用场景来说明“本体”这个术语非常容易。这样的介绍会带来对“本体”术语的操作性理解，使初学者能够识别典型的本体实例，并识别本体可能被用来解决问题的典型情况。（类似于爱德华二世国王宫廷成员对“英寸”的操作性理解。）因此，“本体”的任何定义都只有在我们对定义项的理解超过这种“本体”的操作性理解水平时才可能是有成效的。然而，“概念”和“概念化”这两个术语

¹[1] 物理学IV.213a11-217b28，论灵魂II.418b21-418b26。

length differences below the size of a barley corn. In addition, E1.b enables better measurement precision, since it avoids measurement uncertainty due to variation in barley corn size.

However, imagine some visionary natural philosopher would have proposed to Edward II to use E1.b instead of E1.a. Since Aristotle discussed the existence of a vacuum and the question of the speed of light¹, these concepts would be familiar to scholars of the time. E.g., in the 13th century Witelo considered the possibility that light is faster when traveling through a vacuum than when traveling through a thicker medium [14]. Thus, assuming some Arab mathematician would have helped with the Arabic numerals and fractions, E1.b would not have been outside the conceptual realm of contemporaries of Edward II. But should Edward II have chosen E1.b instead of E1.a?

Within an agricultural medieval society nobody possessed precise clocks, let alone the technology to determine how far light travels in a given fraction of a second. Thus, the contemporaries of King Edward II would not have been able to apply E1.b and use it to determine the length of an object. Therefore, if Edward II would have embraced E1.b, this decision would have had no impact on the actual use of the term “inch”. Since the inch was already established as a measurement unit, people would have continued to use “inch” in the way they always understood it, while paying lip-service to the definition. Hence, the royal definition would not have contributed to a standardization of length measurements. In contrast, E1.a provided a definition of “inch” that was easily accessible and usable for people in an agricultural society.

This example illustrates that the fertility of a definition depends on the context of its use. *Fertility* denotes the degree to which people who embrace a definition are able to utilize it, e.g., to classify some phenomenon or solve a problem or to make some other tangible intellectual process. In the context of the 21st century E1.b is more fertile than E1.a, because it enables better measurements of lengths and connects it systematically to other units of measurement in the International System of Units. However, in the context of a medieval society E1.b would serve no useful function at all; it would be *barren*. This is the case, because King Edward’s contemporaries already had a better understanding of “inch” (the definiendum) than of the definiens of E1.b.

Most definitions of “ontology” are as barren as E1.b would have been for King Edward II, and for the same reasons. It is quite easy to introduce the term “ontology” to a novice to the field by illustrating it with examples and use cases. Such an introduction leads to an operational understanding of the term “ontology”, which allows the novice to recognize typical examples of ontologies and to recognize typical situations where ontologies may be used to solve a problem. (Analogously to the members of King Edward’s II court, who had an operational understanding of “inch”.) Thus, any definition of “ontology” may only be fertile, if our understanding of the definiens exceeds this kind of operational level of understanding of “ontology”. However, the terms “concepts”

¹[1] Physics IV.213a11-217b28, On the Soul II.418b21-418b26.

notoriously很难解释。事实上，“概念”的本质在哲学中是一个悬而未决的问题 [13, 19]。这体现在应用本体文献中对“概念化”的多种解释中，包括：

- 1. “某个领域中假定存在的对象、概念和其他实体，以及它们之间的关系” [6]
- 2. “[...] 为了某种目的而希望表示的世界的一种抽象、简化的视图” [5]
- 3. “一种编码现实结构隐含规则的意向性语义结构” [9]
- 4. “人们用来思考和交流世界的一部分世界的结构化解释” [3]
- 5. “通过识别现象的相关概念，对世界中某种现象的抽象模型” [24]

因此，在“概念化”这一术语普遍模糊的理解背景下，任何基于它的“本体”定义，都像14世纪基于光速的“英寸”定义一样贫瘠。这样的定义可以作为一种学术装饰，但它不会起到解释作用。²

3 迈向“本体”的丰产性定义

本节列举了丰产性的“本体”定义应当能够阐明的一系列现象。

3.1 本体、版本与公式集

根据 Guarino 等人的观点，本体是满足特定条件的逻辑理论——即公式集合—— [11]。毫无疑问，本体涉及逻辑理论，但将本体简单地等同于公式集合是有原因的。首先，本体会随时间变化，因此同一本体的不同版本可能与不同的公式集相关联（例如，因为添加了额外的公理）。此外，同一本体可能用不同的语言维护（例如，OWL 和 FOL）。甚至不能将本体版本等同于公式集合，因为不同的本体版本可能在注释方面有所不同（例如，如果添加了术语的自然语言定义）。任何关于“本体”的解释都需要阐述本体、其版本和逻辑理论之间的关系³

²一项旨在解决此问题并借助形式理论阐明“概念化”的显著尝试最初由 Guarino 在 [8] 提出，并在 [11] 中得到进一步发展。参见第 5 节。³Barry Smith 在一次对话中向我指出了这一事实。

and “conceptualization” are notoriously hard to explain. Indeed, the nature of concepts is an open question in philosophy [13, 19]. This is reflected by the diverse explanations of “conceptualization” in the applied ontology literature, which include:

- 1. “the objects, concepts, and other entities that are assumed to exist in some area of interest and the relationships that hold among them” [6]
- 2. “[...] abstract, simplified view of the world that we wish to represent for some purpose” [5]
- 3. “an intensional semantic structure which encodes the implicit rules constraining the structure of a piece of reality” [9]
- 4. “a structured interpretation of a part of the world that people use to think and communicate about the world” [3]
- 5. “an abstract model of some phenomenon in the world by having identified the relevant concepts of that phenomenon” [24]

Thus, in the context of the prevalent fuzzy understanding of the term “conceptualization”, any definition of “ontology”, which is based on it, is as barren as a definition of “inch” based on the speed of light in the 14th century. Such a definition may serve as a scholarly ornament, but it will not fulfill an explanatory function.²

3 Towards a Fertile Definition of “Ontology”

This section enumerates a number of various phenomena that a fertile definition of “ontology” should be able to illuminate.

3.1 Ontologies, Versions, and Sets of Formulas

According to Guarino et al. ontologies are logical theories – i.e., sets of formulas – that meet certain conditions [11]. Undoubtedly, ontologies involve logical theories, but there are reasons not to simply identify an ontology with a set of formulas. First of all, ontologies change over time and, thus, different versions of the same ontology may be associated with different sets of formulas (e.g., because additional axioms have been added). Further, the same ontology may be maintained in different languages (e.g., OWL and FOL). One cannot even identify an ontology version with a set of formulas, because different ontology versions may differ with respect to their annotations (e.g., if a natural language definition of a term is added). Any explanation of “ontology” needs to address the relationships between ontologies, their versions, and logical theories.³

²A notable attempt to address this issue and clarify “conceptualization” with the help of a formal theory was initially proposed by Guarino in [8] and further developed in [11]. See Section 5.

³This fact was pointed out to me by Barry Smith in a conversation.

3.2 词汇表和注释的重要性

从逻辑角度来看，E.2a-E.2d 之间的逻辑理论没有显著差异，因为对于满足其中之一的任何 OWL 解释，都存在满足其他理论的同构解释。

示例 2

E.2a 类：P 与 R 不相交

E.2b 类：Dog 与 GermanShepherd 不相交

E.2c 类：Biolgy 与 Chenistry 不相交

E.2d 类：Biology 与 Chemistry 不相交

从本体论的角度来看，E.2a-E.2d 差异极大。E.2a 是一个逻辑理论，但它并不代表任何关于某个领域的知识。因此，E.2a 不是本体。相比之下，在没有额外信息的情况下，我们可以合理地假设 E.2b 是一个关于狗的本体，而本体 E.2c 和 E.2d 是关于两个科学之间的关系。唯一的不同是 E.2c 包含拼写错误，而 E.2d 没有。

请注意，从逻辑角度来看，E.2d 与 E.2c 的关系与 E.2b 完全相同。然而，对我们来说显然是 E.2d 和 E.2b 关于不同的领域，而我们在没有额外信息的情况下假设——至少——E.2d 和 E.2c 关于同一个领域。此外，我们知道公理 E.2b 是错误的，而 E.2c 的内容是真实的，但词汇的拼写需要修正。

我们能够做出这些区分，并非因为不同的逻辑理论 E.2a-E.2d 具有任何逻辑属性，而是因为词汇选择是本体论的一个基本组成部分。通过使用某种自然语言的术语，我们确立了本体论的预期解释，并间接地确定了本体论所涉及的领域。如果我们未能建立这种预期解释，例如在 E.2a 的情况下，那么就没有本体论。

注释提供了另一种建立预期解释的工具。我们以最广泛的意义使用“注释”一词，它包括任何作为本体论一部分、旨在记录其词汇或其逻辑公理的材料。这可能包括自然语言定义、说明、注释、示例、参考文献、链接、图表、图片、音频文件或视频文件。

注释很重要，因为将单个术语作为词汇的一部分使用可能会留下不同的解释和歧义。例如，E.3a的逻辑内容与示例E.2d没有区别。但是，注释澄清了术语“生物学”并非指生物学本身，而是指关于生物学的类。请注意，E.3a中的注释包含在公理中未捕获的信息：第一个注释提供了一个标签

3.2 The Importance of the Vocabulary and Annotations

From a logical point of view there is no significant difference between the logical theories E.2a-E.2d, because for any OWL interpretation that satisfies one of them, there are isomorphic interpretations that satisfy the others.

Example 2

E.2a Class: P DisjointWith: R

E.2b Class: Dog DisjointWith: GermanShepherd

E.2c Class: Biolgy DisjointWith: Chenistry

E.2d Class: Biology DisjointWith: Chemistry

From an ontological point of view E.2a-E.2d could not be more different. E.2a is a logical theory, but it is not representing any knowledge about any domain. Thus, E.2a is not an ontology. In contrast, in the absence of additional information, it is reasonable to assume that E.2b is an ontology that is about dogs, and the ontologies E.2c and E.2d are about the relationship of two sciences. The only difference is that E.2c contains spelling errors, while E.2d does not.

Note that, from a logical point of view, E.2d stands in exactly the same relationship to E.2c as to E.2b. Nevertheless, it seems obvious to us that E.2d and E.2b are about different domains, while we assume – at least in the absence of additional information – that E.2d and E.2c are about the same domain. Further, we know that axiom E.2b is false, while the content of E.2c is true but the spelling of the vocabulary needs to be fixed.

We are able to make these distinctions not because of any logical properties of the different logical theories E.2a-E.2d, but because the choice of the vocabulary is an essential component of an ontology. By using terms from a given natural language we establish the *intended interpretation* of the ontology and, indirectly, the domain that the ontology is about. If we fail to establish such an intended interpretation as in the case of E.2a, there is no ontology.

Annotations provide another tool for establishing intended interpretations. We use the term “annotation” in the broadest possible sense that includes any kind of material that is included in an ontology with the intent to document its vocabulary or its logical axioms. This may include natural language definitions, explanations, comments, examples, references, links, diagrams, pictures, audio files or video files.

Annotations are important, because the use of individual terms as part of a vocabulary may leave room for different interpretations and ambiguity. E.g., the logical content of E.3a does not differ from example E.2d. However, the annotations clarify that the term “biology” is not intended to refer to biology, but to classes about biology. Note that the annotations in E.3a contain information that is not captured in the axiom: the first annotation provides a label

对于该类，第二个属性将其公理归属于其创建者，第三个包含一条警告。

这些示例说明了为什么本体（ontology）的公理通常仅反映注释提供信息的一小部分。我们需要区分断言性注释（assertive annotations）和其他注释。断言性注释是指旨在断言关于本体（ontology）领域内真命题的注释。例如，向类添加新标签不涉及断言。元数据（例如，作者信息）的包含是断言，但关于本体本身，而非本体（ontology）的领域。因此，E.3a 的第二个注释不是断言性注释。第三个注释是一个警告。这个特定的警告是一种言语行为，涉及关于本体的断言，即对于某些材料需要得到物质上的家长批准。因此，E.3a 的最后两个注释是断言性的。

本体的逻辑理论可能仅表示断言性注释的内容。E.3a 的第三个注释说明了为什么断言性注释通常没有被公理化。这可能是出于多种原因。例如，因为形式化需要过度扩展词汇表，或者形式语言的表达能力不足。在任何情况下，将本体（ontology）视为仅仅是逻辑理论这种还原论观点，容易忽略它们包含的大量信息。

示例 3

E.3a 类：生物学 DisjointWith：化学 注释：rdfs:label “中学生物课程”，创建者 Fabian, rdfs:comment “警告：涉及需要家长批准的内容”，rdfs:comment “没有生物课程是化学课程”

E.3b 类：P 注释：rdfs:label “狗” DisjointWith：R 类：

R 注释：rdfs:label “德国牧羊犬”

E.3c 类：狗 DisjointWith：德国牧羊犬 注释：rdfs:comment “德国牧羊犬是一种受欢迎的狗品种”

示例 E.3b 说明注释足以建立预期解释，即使词汇由任意符号组成。请注意，E.3b 的逻辑理论与 E.2a 相同，但在注释的帮助下，理论被转化为本体。实际上，有人可以论证 E.3a 和 E.2b 在本体上是等价的，尽管它们的逻辑理论显然不是逻辑等价的。

示例 E.3c 显示，逻辑理论与本体的注释之间可能存在不匹配。由于逻辑理论是一致的，因此无法借助 OWL 推理器来检测这类问题，

for the class, the second attributes the axiom to its creator, the third includes a warning.

These examples illustrate why, typically, the axioms of an ontology reflect only a fraction of the information that is provided by the annotations. We need to distinguish assertive annotations from the other annotations. *Assertive annotations* are the kind of annotations that are intended to assert a true proposition about the domain of the ontology. E.g., adding a new label to a class does not involve an assertion. The inclusion of metadata (e.g., authorship) is an assertion, but about the ontology and not about the domain of the ontology. Thus, the second annotation of E.3a is not an assertive annotation. The third annotation is a warning. This particular warning is a speech act that involves an assertion about the domain, namely, that for some material parental approval is required. Hence, the last two annotations of E.3a are assertive.

The logical theory of an ontology may only represent the content of assertive annotations. The third annotation of E.3a illustrates that often some assertive annotations are not axiomatized. This may be the case for various reasons. E.g., because the formalization would require an excessive extension of the vocabulary or the formal language is not expressive enough. In any case, a reductionistic view on ontologies that treats them as mere logical theories is prone to ignore much of the information they contain.

Example 3

*E.3a Class: Biology DisjointWith: Chemistry
Annotations: rdfs:label “Biology classes for middle school”, creator Fabian,
rdfs:comment “Warning: involves material that requires parental approval”,
rdfs:comment “No biology classes are chemistry classes”*

*E.3b Class: P Annotations: rdfs:label “Dog” DisjointWith: R
Class: R Annotations: rdfs:label “German Shepherd”*

*E.3c Class: Dog DisjointWith: GermanShepherd
Annotations: rdfs:comment “German Shepherd is a popular breed of dog”*

Example E.3b illustrates that annotations are sufficient to establish an intended interpretation, even if the vocabulary consists of arbitrary symbols. Note that the logical theory of E.3b is identical to E.2a, but with the help of the annotations the theory is turned into an ontology. Indeed, one could argue that E.3a and E.2b are *ontologically equivalent*, although their logical theories are clearly not logically equivalent.

Example E.3c shows that there may be a *mismatch* between the logical theory and the annotations of an ontology. It is not possible to detect this kind of problem with the help of an OWL reasoner, since the logical theory is

问题产生的原因是注释与注释相互矛盾。

前面的例子表明，词汇表和注释的选择至关重要，因为它们与逻辑理论相结合，使用户能够建立本体的预期解释。我们所说的后者是指用户能够识别本体的领域，并理解注释和逻辑理论所断言的关于该领域的命题。与任何复杂文本一样，理解本体通常是一个解释过程，它从对本体非常初步的理解开始，最终达到能够理解整体概览以及所有细节的阶段。这种过程的成功不仅取决于本体的质量，还取决于用户的知识。例如，如果词汇表和注释使用的是英语，那么对于不懂英语的用户来说，它们的作用将非常有限。一个良好的“本体”定义应该反映出词汇表和注释的选择至关重要。此外，它还应该能够解释公理与注释之间的本体等价性和不匹配。

3.3 分享一个本体

从 Gruber [5], 开始，本体在知识共享中的作用在关于本体性质的讨论中尤为突出。本体的两个重要优势在于，它可以作为不同应用的知识资源，并且本体使不同的人（和系统）能够交换信息。因此，本体通常被定义为一种规范，它指定了一个“共享概念化”（见第 5 节）。这存在两个问题。

首先，如果假设概念化是一种基于经验构建的领域心理表征，就会产生一个哲学挑战。正如 [11] 指出的，在这种假设下，很难从字面意义上理解概念化是如何被共享的。因为心理实体在意义上必然是私有的，因为它们只对体验它们的人是可访问的。

其次，本体基于共享概念化的说法，与大型本体开发中典型的知识分工相矛盾。想象一下，一家手机制造商开发本体，目的是在其产品的整个生命周期中体现关于产品的知识。因此，本体需要为任何手机代表其设计（形状、尺寸、触觉特性）、其电子部件及其功能、组件的拓扑结构、手机装配计划，以及关于维修和回收的信息。显然，这些方面中的每一个都由专家开发。假设本体开发团队由来自各个领域的专家组成。通常，手机的设计师对装配计划知之甚少，而负责回收的人员通常不会了解部件的功能规格细节。因此，本体开发团队中任何成员对手机领域的概念化在其专业领域内非常详细，在其他领域则很浅。因此，本体的领域概念化在其开发者之间差异很大。因此，没有单一的领域概念化被所有人共享。实际上，由于本体包含的信息比任何单个团队成员所拥有的都多，因此本体没有指定任何个人的概念化。

logically consistent. The problem arises because the annotation contradicts the annotation.

The previous examples show that the choice of the vocabulary and the annotations play an important role, since in tandem with the logical theory they allow the user of the ontology to establish the intended interpretation of an ontology. By the latter we mean that the user is able to identify the domain of the ontology and to understand the propositions that are asserted about the domain by the annotations and the logical theory. As it is the case for any complex texts, understanding an ontology is typically a hermeneutic process, which leads from a very preliminary understanding of the ontology to a situation where one understands the big picture as well as all the details. The success of such a process depends not only on the quality of the ontology, but also on the knowledge of the user. If, for example, the vocabulary and the annotations are in English, they will be of limited use to a user who does not speak English. A good definition of “ontology” should reflect that the choice of the vocabulary and the annotations play an important role. Further, it should enable an explanation of ontological equivalence and mismatch between axioms and annotations.

3.3 Sharing an Ontology

Starting with Gruber [5], the role of ontologies for sharing knowledge featured prominently in discussions on the nature of ontologies. Two important benefits of an ontology are that it may be used as a knowledge resource for different applications, and that the ontology enables different people (and systems) to exchange information. For this reason an ontology is often defined as a specification that specifies a “shared conceptualization” (see Section 5). This is problematic for two reasons.

Firstly, a philosophical challenge arises if one assumes that a conceptualization is a kind of mental representation of a domain that is built based on experiences. As [11] points out, under this assumption it is hard to see how conceptualizations may be shared in the literal sense of the word. Because mental entities are necessarily private in the sense that they are not accessible to anyone except the person who experience them.

Secondly, the claim that ontologies are based on a shared conceptualization contradicts the *intellectual division of labor* that is typical for the development of large ontologies. Imagine that a manufacturer of cell phones develops an ontology in order to represent knowledge about its products across their whole life-cycles. Thus, the ontology needs to represent, for any phone, its design (form, size, haptic features), its electronic parts and their functions, the topology of the components, a plan of how the phone is assembled, and information about repair and recycling. Obviously, every of these aspects is developed by specialists. Let’s assume that the ontology development team consists of experts from the various areas. Typically, the designers of the phone have little

手机领域的概念化由本体开发团队中的任何成员在其专业领域内非常详细，在其他领域则很浅。因此，本体的领域概念化在其开发者之间差异很大。因此，没有单一的领域概念化被所有人共享。实际上，由于本体包含的信息比任何单个团队成员所拥有的都多，因此本体没有指定任何个人的概念化。

虽然本体论的开发者通常会对领域持有不同的概念化理解，但必须存在一定的协调，否则本体的开发将会失败。我们该如何解释这一点？

3.4 概念主义与实在论

本体实在论者反对将“本体”定义为一种概念化规范，因为他们认为——根据他们的观点——本体学家关注的是现实而非概念化 [21]。这里不是重述本体实在论者与概念主义者之间辩论论点的合适场所。然而，由于两种立场都有其合理性，因此似乎有必要以一种与两种立场都兼容的方式定义“本体”，并且理想情况下能够阐明它们之间的确切差异。

4 什么是本体？

4.1 本体是文档

本体是一种文档，就像一本书或一个会议日程。由于本体是文档，因此适用于其他文档的系统性歧义同样适用于本体。例如，在句子“会议日程在我们上次的规划会议中得到了完善”、“我昨天下载了会议日程”和“你会在你的会议材料中找到一个会议日程”中，“会议日程”这个表达指的是三个不同的实体。在第一句中，会议日程是一个可能随时间演变的实体，第二句是关于第一个实体的特定快照或版本，第三句是关于第二个实体的特定打印件（或符号）。同样的歧义也适用于本体，例如，考虑“基因本体自2003年以来发生了显著变化。我昨天下载了它，它在我的笔记本电脑的家目录中。”

这类系统性的歧义是良性的，我们已经如此习惯于它们，以至于通常不会注意到。但鉴于本文的目标是对“本体”一词进行明确定义，避免歧义是值得的。因此，为了本文的目的，我们将文档和文档版本区分开来。

knowledge about the assembly plan, while the people who are responsible for recycling typically won’t know details about the functional specifications of the parts. Hence, the conceptualization of the cell phone domain by any member of the ontology development team is very detailed in the area that the person specializes in, and shallow in other areas. Thus, the conceptualizations of the domain of the ontology varies significantly across its developers. Therefore, there is no single conceptualization of the domain that is shared by everybody. Indeed, since the ontology will contain more information than any individual team member possesses, the ontology specifies no single person’s conceptualization.

While the developers of an ontology typically will possess different conceptualizations of the domain, there must be some harmonization, otherwise the development of the ontology will fail. How do we explain this?

3.4 Conceptualism vs. Realism

Ontological realists object against any definition of “ontology” as a specification of a conceptualization because – according to them – ontologists are not about conceptualizations but about reality [21]. This is not the place to recapitulate the arguments in the debate between ontological realists and conceptualists. However, since either position is reasonable, it seems desirable to define “ontology” in a way that is compatible with either position and, ideally, clarifies the exact difference between them.

4 What is an Ontology?

4.1 Ontologies are Documents

An ontology is a kind of document, like a book or a schedule of a conference. Because ontologies are documents, the same kind of systematic ambiguities that apply to other documents apply to ontologies. E.g., in the sentences “The conference schedule was fleshed out in our last planning meeting” and “I downloaded the conference schedule” and “You will find a conference schedule among your meeting materials”, the expression “conference schedule” denotes three different entities. In the first sentence the conference schedule is an entity that may evolve over time, the second sentence is about a particular snapshot or version of the first entity, and the third is about particular printouts (or tokens) of the second entity. The same ambiguities apply to ontologies, e.g., consider “The Gene Ontology changed significantly since 2003. I downloaded it yesterday, it is in my home directory on my laptop.”

These kind of systematic ambiguities are benign and we are so used to them that we usually do not notice. But given that the goal of this paper is a clear definition of the term “ontology”, it is worth to avoid ambiguity. Thus, for the purpose of this paper we distinguish between documents and document ver-

这种区分遵循联合国系统文档本体（UNDO）[20]，根据该本体，一个文档由其版本实现。一个文档的特定版本可能以不同的方式从另一个版本中派生出来。UNDO 区分修订、翻译和转换。

文档的一般区分也适用于本体。本体版本是本体的一个实现。本体版本可以通过多种方式从先前版本派生出来，例如，通过添加新的公理、删除注释或纠正拼写错误。此外，本体版本可能是将注释翻译成不同语言（例如，从英语翻译成西班牙语）的结果，也可能是将逻辑理论从一种逻辑翻译成另一种逻辑（例如，从OWL翻译成一阶逻辑），或者通过改变逻辑理论的序列化方式（例如，从OWL曼彻斯特语法序列化改为Turtle）。因此，本体版本并非孤立存在，而是形成一个网络。这个网络可以被视为一个有向无环图，其中边表示衍生自关系。这引出了定义1。

定义1（本体）一个领域的本体是一个由关于该领域的本体版本网络实现的文档。

当然，这个定义仅将我们最初的问题向前推进了一步，我们需要问自己：关于某个领域的本体版本是什么？

4.2 本体版本

如我们在第3.2节所见，一个本体版本涉及三个元素：词汇表、注释以及某个逻辑中的逻辑理论；而词汇表和注释在建立本体的预期解释中起着重要作用。对于任何试图更深入分析这些观察结果的人来说，一个挑战在于必须避免对所涉及实体过于具体的假设。词汇表很可能由字符串组成，但即使如此也不一定如此。如上所述，注释有多种形式，其中一些是断言性的，而另一些则不是。而且，由于本体是用各种逻辑形式书写的，我们也不能对逻辑做出任何强假设。最重要的是，需要解决这个相当神秘的“预期解释”的性质。

让我们从基础开始。通过感兴趣的领域 $\mathbb{D}$ ，我们理解是一个或多人可能感兴趣的主体。例如包括量子物理学、利物浦FC、基数和权力的游戏。

通过命题，我们理解断言性句子的语义内容。命题可能是真的或假的，并且它们是信念的客体。似乎有理由假设，像“约翰秃头而且弗雷德很高”这样的复杂句子所断言的命题，在某种程度上是结构化的，并且是相互关联的

⁴它们与文档标记（例如，硬盘上的特定文件或特定的打印输出）有所区别。文档标记与本文的目的无关。

sions⁴. This distinction follows the United Nations System Document Ontology (UNDO) [20], according to which a document is realized by its versions. A particular version of a document may be derived from another version in different ways. The UNDO distinguishes between revisions, translations, and transformations.

The distinctions for documents in general also apply to ontologies. An *ontology version* is a realization of an ontology. An ontology version may be derived from a previous version in various ways, e.g., by adding new axioms, removing annotations, or correcting spelling mistakes. Further, an ontology version may be the result of translating the annotations in a different language (e.g., from English to Spanish) or by translating the logical theory from one logic to another (e.g., from OWL to first-order logic) or by changing the serialization of the logical theory (e.g., from a serialization in OWL Manchester Syntax to Turtle). Thus, ontology versions do not exist in isolation, but form a network. This network can be thought of as a directed acyclic graph where the edges represent the derives-from-relationship. This leads us to Definition 1.

Definition 1 (Ontology) *An ontology of a domain is a document that is realized by a network of ontology versions about the domain.*

Of course, this definition only pushes our original question one step further and we need to ask ourselves: *What is an ontology version about a domain?*

4.2 Ontology versions

As we have seen in Section 3.2, an ontology version involves three elements: a vocabulary, annotations, and a logical theory in some logic; and the vocabulary and the annotations play an important role in establishing the intended interpretation of an ontology. For anyone who attempts to analyze these observations closer, one challenge is that one has to avoid too specific assumptions about the entities involved. The vocabulary most likely consists of strings, but even that is not necessarily the case. As discussed above, annotations come in a wide variety of forms, some of them are assertive and some are not. And since ontologies are written in various logical formalisms, we cannot make any strong assumptions about the logic either. Most importantly, one needs to address the nature of the rather mystical ‘intended interpretation’.

Let’s start with the basics. By a *domain of interest* $\mathbb{D}$ we understand a topic that one or more people may be interested in. Examples include quantum physics, FC Liverpool, cardinal numbers, and Game of Thrones.

By *propositions* we understand the semantic contents of assertive sentences. Propositions may be true or false, and they are subjects of beliefs. It seems plausible to assume that the proposition that is asserted by a complex sentence like “John is bald and Fred is tall” is structured in some way and is connected

⁴Both are distinguished from document tokens (e.g., particular files on a hard drive or particular printouts). Document tokens are not relevant for the purpose of this paper.

在某些有趣的方式上与“约翰秃头”所断言的命题相关。然而，在本文的语境中，我们抽象了命题的内部结构，并不担心它们如何相互关联。唯一的例外是我们假设任何命题集要么是一致的，要么是不一致的。两个命题集 $\mathbb{P}_1, \mathbb{P}_2$ 弱等价当且仅当，对于任何命题集 $\mathbb{P}_3$ 、 $\mathbb{P}_1 \cup \mathbb{P}_3$ 是不一致的当且仅当 $\mathbb{P}_1 \cup \mathbb{P}_3$ 是不一致的。对于任何给定的领域 $\mathbb{D}$ ， $\mathbb{P}_{\mathbb{D}}$ 是关于 $\mathbb{D}$ 的命题集。假设 $P^\dagger \subseteq \mathbb{P}_{\mathbb{D}}$ 。一个一致的命题集 $P \subseteq \mathbb{P}_{\mathbb{D}}$ 是 $P^\dagger$ 的极大一致子集，当且仅当不存在一致命题集 P' 使得 $P \subset P'$ 并且 $P' \subseteq P^\dagger$ 。

既然我们不想为了我们的目的而考虑特定的逻辑，我们将逻辑定义为一个蕴涵系统。⁵

定义 2（逻辑）逻辑 $L = \langle \text{Sym}, \text{Sen}, \text{sign} \vdash \rangle$ 由组成

1. 符号集 Sym
2. 句子集 Sen over Sym
3. 函数签名 $:\wp(\text{Sen}) \rightarrow \wp(\text{Sym})$ ，它将一组句子映射到其签名（即，其中出现的符号的集合），使得
 - (a) $\text{sign}(/0) = /0$ 和 $\text{sign}(\text{Sen}) = \text{Sym}$ ；
 - (b) 如果 $\Gamma \subseteq \Delta \subseteq \text{Sen}$ ，那么 $\text{sign}(\Gamma) \subseteq \text{sign}(\Delta)$
4. 一个蕴涵关系 $\vdash \subseteq \wp(\text{Sen}) \times \text{Sen}$ 。

根据我们对逻辑的定义，我们可以引入带注解的逻辑理论的概念。对于任何逻辑 $L = \langle \text{Sym}, \text{Sen}, \text{sign} \vdash \rangle$ ， Γ 是 L 中的一个理论，当且仅当 $\Gamma \subseteq \text{Sen}$ ； Γ 是 L 中关于 V 的一个理论，当且仅当 $V \subseteq \text{Sym}$ 并且 Γ 是 L 中的一个理论，并且 $\text{sign}(\Gamma) \subseteq V$ 。

定义 3（带注解的逻辑理论）一个带注解的逻辑理论 $\langle V, \Gamma, \text{Anno}, L \rangle$ 由以下部分组成：

1. 一个非空的符号集合 V （词汇表）；
2. 一个 L 上的关于 V 的理论 Γ ；
3. 一个注释集合 Anno ，它被划分为断言性注释集合 Anno^a 和非断言性注释集合 Anno^n ；
4. 逻辑 L 。

给定一个关于某个领域的断言性语句集合，其预期解释可以被视为将这些句法实体映射到关于该领域的命题。在本体的情况下，这些断言性语句有两种类型：断言性注释和本体的公理。

⁵正如 [2] 所指出的，为了包含线性逻辑等形式化系统，就必须使用多重集而不是集合。

定义2比 [2] 中的对应定义更丰富，因为选定的词汇表对本体论很重要。另一个区别是，定义2允许前提集是无限的。

in some interesting fashion to the proposition that is asserted by “John is bald”. However, within the context of this paper we abstract from the internal structure of propositions and do not worry about how they relate to each other. The only exception is that we assume that any set of propositions is either *consistent* or *inconsistent*. Two sets of propositions $\mathbb{P}_1, \mathbb{P}_2$ are *weakly equivalent* iff, for any set of propositions $\mathbb{P}_3$, $\mathbb{P}_1 \cup \mathbb{P}_3$ is inconsistent iff $\mathbb{P}_1 \cup \mathbb{P}_3$ is inconsistent. For any given domain $\mathbb{D}$, $\mathbb{P}_{\mathbb{D}}$ is the set of propositions that is about $\mathbb{D}$. Assume $P^\dagger \subseteq \mathbb{P}_{\mathbb{D}}$. A consistent set of propositions $P \subseteq \mathbb{P}_{\mathbb{D}}$ is a *maximal consistent* subset of $P^\dagger$ iff there is no consistent set of propositions P' such that $P \subset P'$ and $P' \subseteq P^\dagger$.

Since we do not want to consider a particular logic for our purpose, we define a logic as an entailment system.⁵

Definition 2 (Logic) A logic $L = \langle \text{Sym}, \text{Sen}, \text{sign}, \vdash \rangle$ consists of

1. the set of symbols **Sym**;
2. the set of sentences **Sen** over **Sym**;
3. the function **sign** : $\wp(\text{Sen}) \rightarrow \wp(\text{Sym})$ that maps a set of sentences to its signature (i.e., the set of symbols that occur in it), such that
 - (a) **sign**($\emptyset$) = $\emptyset$ and **sign**(**Sen**) = **Sym**;
 - (b) if $\Gamma \subseteq \Delta \subseteq \text{Sen}$, then **sign**(Γ) $\subseteq$ **sign**(Δ);
4. an entailment relationship $\vdash \subseteq \wp(\text{Sen}) \times \text{Sen}$.

Based on our definition of a logic, we may introduce the notion of an annotated logical theory. For any logic $L = \langle \text{Sym}, \text{Sen}, \text{sign}, \vdash \rangle$, Γ is a theory in L iff $\Gamma \subseteq \text{Sen}$; Γ is a theory over V in L iff $V \subseteq \text{Sym}$ and Γ is a theory in L and **sign**(Γ) $\subseteq V$.

Definition 3 (Annotated Logical Theory) An annotated logical theory $\langle V, \Gamma, \text{Anno}, L \rangle$ consists of

1. a non-empty set of symbols **V** (the vocabulary);
2. a theory Γ over **V** in L ;
3. a set **Anno** of annotations, which is partitioned into the set of assertive annotations **Anno**^a and non-assertive annotations **Anno**ⁿ;
4. a logic L .

Given a set of assertive statements about a domain, their intended interpretation may be considered as a mapping from these syntactic entities to propositions about the domain. In the case of the ontology these assertive statements come in two flavors: the assertive annotations and the axioms of the ontology.

⁵As [2] points out, in order to include formalisms like linear logic, one would have to use multisets instead of sets. Def. 2 is richer than the corresponding definition in [2], since the chosen vocabulary is important for ontologies. One other difference is that Def. 2 allows the set of premises to be infinite.

定义4 设 $\Lambda = \langle V, \Gamma, \text{Anno}, L \rangle$ 是一个带注释的逻辑理论，设 $\mathbb{D}$ 是一个领域。一个从 Λ 到 $\mathbb{D}$ 的预期解释 $\mathcal{I}$ 是一个映射，满足：(1) 对于任何注释 $a \in \text{Anno}^a$ ，有 $\mathcal{I}(a) \in \mathbb{P}_{\mathbb{D}}$ ；以及(2) 对于任何句子 $\phi \in \Gamma$ ，有 $\mathcal{I}(\phi) \in \mathbb{P}_{\mathbb{D}}$ 。

如我们在第4.1节中讨论的，本体版本是一种文档版本。现在，我们可以更具体地说明：本体版本的作者以一种方式指定了注释理论，使用户能够理解其预期解释（参见定义5）。这通常涉及使用已建立的领域术语，作为词汇表或词汇表的标签。请注意，本体版本不一定需要包含注释逻辑理论的完整词汇表、公理和注释，因为本体版本可能包含导入内容。

定义5（本体版本）设 $\mathbb{D}$ 为一个领域。

$\mathbf{o}$ 是一个关于 $\mathbb{D}$ 的本体版本，当且仅当 $\mathbf{o}$ 是一个指定了恰好一个注释逻辑理论 $\Lambda = \langle V, \Gamma, \text{Anno}, L \rangle$ 的文档版本，使得词汇表 V 的选择连同 Anno 中的注释和 Γ 中的公理能够使本体的合格用户确定预期解释函数 $\mathcal{I}$ 从 Λ 到 $\mathbb{D}$ 。

令 $\mathbb{P}^{\mathbf{o}}$ 是实体 $\mathbf{o}$ 所断言的命题集合，其中 $\mathbb{P}^{\mathbf{o}} = \mathbb{P}^{\mathbf{o}}_{\text{Anno}} \cup \mathbb{P}^{\mathbf{o}}_{\Gamma}$ 和 $\mathbb{P}^{\mathbf{o}}_{\text{Anno}} = \{ p \in \mathbb{P}_{\mathbb{D}} \mid p = \mathcal{I}(a), a \in \text{Anno}^a \}$ 和 $\mathbb{P}^{\mathbf{o}}_{\Gamma} = \{ p \in \mathbb{P}_{\mathbb{D}} \mid p = \mathcal{I}(\phi) \text{ 以及 } \Gamma \vdash \phi \}$ 。

在定义 5 $\mathbb{P}^{\mathbf{o}}_{\text{Anno}}$ 中，是本体 $\mathbf{o}$ 断言注释的预期解释的命题集合。类似地， $\mathbb{P}^{\mathbf{o}}_{\Gamma}$ 是预期解释由 Γ 推导出的公式的命题集合。定义 5 允许我们定义各种概念。令 $\mathbf{o}_1, \mathbf{o}_2$ 是关于领域 $\mathbb{D}$ 的本体。 $\mathbf{o}_1, \mathbf{o}_2$ 是强等价的，当且仅当 $\mathbb{P}^{\mathbf{o}_1} = \mathbb{P}^{\mathbf{o}_2}$ 。 $\mathbf{o}_1, \mathbf{o}_2$ 是弱等价的，当且仅当 $\mathbb{P}^{\mathbf{o}_1}$ 和 $\mathbb{P}^{\mathbf{o}_2}$ 是弱等价的。 $\mathbf{o}_1$ 比 $\mathbf{o}_2$ 更强，当且仅当存在某个 $\mathbb{P}' \subseteq \mathbb{P}_{\mathbb{D}}$ ，使得 $\mathbb{P}^{\mathbf{o}_1}$ 和 $\mathbb{P}^{\mathbf{o}_2} \cup \mathbb{P}'$ 是弱等价的。 $\mathbf{o}$ 包含逻辑理论和注释之间的不一致，当且仅当 $\mathbb{P}^{\mathbf{o}}_{\text{Anno}}$ 和 $\mathbb{P}^{\mathbf{o}}_{\Gamma}$ 每个都是一致的，但 $\mathbb{P}^{\mathbf{o}}_{\text{Anno}} \cup \mathbb{P}^{\mathbf{o}}_{\Gamma}$ 是不一致的。

这些定义使我们能够解释示例 E.2 和 E.3。E.2a 不是一个本体版本，因为它未能建立预期的解释。E.2b 和 E.2d 的预期解释将它们的公理映射到不同的命题。相比之下，E.2c 和 E.2d 被映射到同一个命题，因此它们在本体上是强等价的。与 E.2a 不同的是，E.3b 中的注释指定了预期的解释。E.3b 和 E.2b 在本体上是强等价的。E.3c 包含其注释与其逻辑理论之间的不匹配。

定义5并不假设整个文档是用某种形式语言编写的。我们不包含这一要求，因为已知有许多本体出版物并不满足该要求。例如，在 [15] DOLCE是在一个包含自由文本和公理的PDF文档中指定的。当然，公理是用形式语言书写的，但整个文档不是。

Definition 4 Let $\Lambda = \langle V, \Gamma, \text{Anno}, L \rangle$ be an annotated logical theory and let $\mathbb{D}$ be a domain. An intended interpretation $\mathcal{I}$ from Λ into $\mathbb{D}$ is a mapping such that: (1) for any annotation $a \in \text{Anno}^a$, $\mathcal{I}(a) \in \mathbb{P}_{\mathbb{D}}$; and (2) for any sentence $\phi \in \Gamma$, $\mathcal{I}(\phi) \in \mathbb{P}_{\mathbb{D}}$.

As we have discussed in Section 4.1, an ontology version is a kind of document version. Now, we can be more specific: the authors of an ontology version specify an annotated theory in such a way that the users are able to grasp its intended interpretation (see Definition 5). This usually involves the use of an established terminology for the domain either as the vocabulary or as labels for the vocabulary. Note that the ontology version does not necessarily need to contain the complete vocabulary, axioms, and annotations of the annotated logical theory, since the ontology version may contain importations.

Definition 5 (Ontology Version) Let $\mathbb{D}$ be a domain.

$\mathbf{o}$ is an ontology version about $\mathbb{D}$ iff $\mathbf{o}$ is a document version that specifies exactly one annotated logical theory $\Lambda = \langle V, \Gamma, \text{Anno}, L \rangle$ such that the choice of the the vocabulary V together with the annotations in Anno and the axioms in Γ enable a competent user of the ontology to ascertain the intended interpretation function $\mathcal{I}$ from Λ into $\mathbb{D}$.

Let $\mathbb{P}^{\mathbf{o}}$ be the set of propositions asserted by $\mathbf{o}$, where $\mathbb{P}^{\mathbf{o}} = \mathbb{P}^{\mathbf{o}}_{\text{Anno}} \cup \mathbb{P}^{\mathbf{o}}_{\Gamma}$ and $\mathbb{P}^{\mathbf{o}}_{\text{Anno}} = \{ p \in \mathbb{P}_{\mathbb{D}} \mid p = \mathcal{I}(a), a \in \text{Anno}^a \}$ and $\mathbb{P}^{\mathbf{o}}_{\Gamma} = \{ p \in \mathbb{P}_{\mathbb{D}} \mid p = \mathcal{I}(\phi) \text{ and } \Gamma \vdash \phi \}$.

In Definition 5 $\mathbb{P}^{\mathbf{o}}_{\text{Anno}}$ is the set of propositions that are the intended interpretations of the assertive annotations of the ontology $\mathbf{o}$. Analogously, $\mathbb{P}^{\mathbf{o}}_{\Gamma}$ is the set of propositions that are intended interpretations of formulas that are entailed by Γ . Definition 5 allows us to define various notions. Let $\mathbf{o}_1, \mathbf{o}_2$ be ontologies about a domain $\mathbb{D}$. $\mathbf{o}_1, \mathbf{o}_2$ are *strongly equivalent* iff $\mathbb{P}^{\mathbf{o}_1} = \mathbb{P}^{\mathbf{o}_2}$. $\mathbf{o}_1, \mathbf{o}_2$ are *weakly equivalent* iff $\mathbb{P}^{\mathbf{o}_1}$ and $\mathbb{P}^{\mathbf{o}_2}$ are weakly equivalent. $\mathbf{o}_1$ is *stronger* than $\mathbf{o}_2$ iff there is some $\mathbb{P}' \subseteq \mathbb{P}_{\mathbb{D}}$ such that $\mathbb{P}^{\mathbf{o}_1}$ and $\mathbb{P}^{\mathbf{o}_2} \cup \mathbb{P}'$ are weakly equivalent. $\mathbf{o}$ contains a *mismatch between the logical theory and the annotations* iff $\mathbb{P}^{\mathbf{o}}_{\text{Anno}}$ and $\mathbb{P}^{\mathbf{o}}_{\Gamma}$ each are consistent, but $\mathbb{P}^{\mathbf{o}}_{\text{Anno}} \cup \mathbb{P}^{\mathbf{o}}_{\Gamma}$ is inconsistent.

These definitions allows us to explain the examples E.2 and E.3. E.2a is not an ontology version, since it does fail to establish an intended interpretation. The intended interpretations of E.2b and E.2d map their axioms to different propositions. In contrast, E.2c and E.2d are mapped to the same proposition, and, thus, they are strongly ontologically equivalent. In contrast to E.2a, the annotations in E.3b specify the intended interpretation. E.3b and E.2b are ontologically strongly equivalent. E.3c contains a mismatch between its annotations and its logical theory.

Definition 5 does not assume that the whole document is written in some formal language. We do not include this requirement, since there are well-known publications of ontologies that do not meet it. E.g., in [15] DOLCE is specified in a PDF document that contains free text and axioms. Of course, the axioms are written in a formal language, but the overall document is not.

根据定义5，本体基本上是一个被注释的逻辑理论，它被解释为关于某个领域的命题。这些命题不需要为真或对应于概念化。因此，即使是一个只包含错误公理的本体也是一个本体。本体如何与概念化或现实相关联将在下文中讨论。

4.3 从实在论者视角看本体

根据实在论，本体是关于现实的一个特定切片（从特定视角、在给定粒度级别上观察）。让我们把符合该描述的领域称为实在领域。斑马鱼的解剖结构和伊朗的政治结构是实在领域的例子。相比之下，独角兽的解剖结构或《权力的游戏》世界中的七大王国的政治结构就不是实在领域。因此，从实在论者的视角来看，这些主题不适合作为本体的研究领域。这与概念论者的立场存在一个主要区别，后者所关心的领域可能涉及虚构实体。

一种表征实在领域的方式是通过该领域中可能发生的不同状态。状态可以用关于该领域的命题的最大一致集来形式化表示。因此，为了本文的目的（诚然，这是一个强烈的简化），我们假设如下：实在领域是 $\mathbb{P}_{\mathbb{D}}$ 的最大一致子集的集合。基于这个定义，我们可以定义何时一个实在领域的本体是正确的：

定义 6 (CorrectOntology) 一个文档 $\mathbf{o}$ 是一个真实领域 $\mathbb{D}$ 的正确本体版本，当且仅当 $\mathbf{o}$ 是关于 $\mathbb{D}$ 的本体版本，并且对于任何事件状态 $s \in \mathbb{D}$ ： $\mathbb{P}^{\mathbf{o}} \subseteq s$ 。

4.4 从概念论者的视角看本体

根据概念主义，本体指定概念化（某个领域的）。如第2节所述，概念化的性质尚不明确。然而，我们将对概念化做出以下假设：如果智能体内化了某个领域 $\mathbb{C}$ 的概念化 $\mathbb{D}$ ，那么当呈现给智能体某个状态 s 时，该概念化将使智能体能针对某个命题 $p \in \mathbb{P}_{\mathbb{D}}$ 判断：(a) 根据 $\mathbb{C}$ ， p 在 s 中为真；(b) 根据 $\mathbb{C}$ ， p 在 s 中为假；(c) p 在 s 中的真值不由 $\mathbb{C}$ 决定。这一假设使我们避免了关于概念化性质的棘手问题。相反，我们将它们视为黑箱，其特征在于内化概念化的智能体能回答问题（或表现出其他允许我们推断智能体如何评估命题的行为）。由于这是一种行为主义转向，我们将这种特征称为给定领域的概念化行为特征（BPC）。此类BPC包含该概念化所涉及领域中的状态集合，并为每个状态提供一个从命题到真或假的偏映射。该映射是偏的，因为我们允许领域的不完整概念化。

According to Definition 5 an ontology is basically an annotated logical theory which is interpreted as propositions about a domain. These propositions do not need to be true or correspond to a conceptualization. Thus, even an ontology that contains only false axioms is an ontology. How ontologies relate to conceptualizations or reality will be discussed next.

4.3 Ontologies from a Realists’ Perspective

According to ontological realism, an ontology is about a particular slice of reality (on a given level of granularity viewed from a particular perspective) [21]. Let’s call a domain of interest that meets that description a real domain. The anatomy of zebrafish and the political structure of Iran are examples for real domains. In contrast, the anatomy of unicorns or the political structure of the Seven Kingdoms in the Game of Thrones world are not real domains. Thus, from a realist’s perspective these topics are no suitable domains of interest for an ontology. This is a major difference to the conceptualist’s position, whose domain of interest may involve fictional entities.

One way to characterize a realistic domain is by the different states of affair that may come about in this domain. A state of affair may be formally represented by maximal consistent sets of propositions about the domain. Thus, for the purpose of this paper (admittedly, a strong simplification) we assume the following: A *real domain* $\mathbb{D}$ is a set of maximal consistent subsets of $\mathbb{P}_{\mathbb{D}}$. Based on this definition, we may define when an ontology of a real domain is correct:

Definition 6 (Correct Ontology) A document $\mathbf{o}$ is a correct ontology version of a real domain $\mathbb{D}$ iff $\mathbf{o}$ is an ontology version about $\mathbb{D}$ and for any state of affair $s \in \mathbb{D}$: $\mathbb{P}^{\mathbf{o}} \subseteq s$.

4.4 Ontologies from a Conceptualist’s Perspective

According to ontological conceptualism, ontologies specify conceptualizations (of some domain). As discussed in Section 2, the nature of conceptualizations is unclear. However, we will make the following assumption about conceptualizations: if an agent internalizes a conceptualization $\mathbb{C}$ of a given domain $\mathbb{D}$, then the conceptualization enables the agent when presented a given state of affairs s to evaluate for a given proposition $p \in \mathbb{P}_{\mathbb{D}}$ whether (a) p is true in s according to $\mathbb{C}$ or (b) p is false in s according to $\mathbb{C}$ or (c) whether the truth-value of p in s is not determined by $\mathbb{C}$. This assumption allows us to avoid the thorny question of the nature of conceptualizations. Instead we treat them as black boxes that are characterized by the ability of an agent who internalizes the conceptualization to answer questions (or show some other behavior that allows us to infer how the agent evaluates propositions). Since this is a kind of behavioristic turn, we call this characterization the *behavioral profile of a conceptualization* (BPC) of a given domain. Such a BPC consists of the set of

状态，并提供每个状态的命题到真或假的偏映射。该映射是偏的，因为我们允许领域的不完整概念化。

定义 7（概念化行为特征 (BPC)）一个领域 $\mathbb{D}$ 的 BPC 是一个有序对 $\langle \mathbb{S}, v \rangle$ ，其中

1. 状态集 $\mathbb{S}$ 非空；
2. 赋值函数 v 为每个状态 $s \in \mathbb{S}$ 分配一个部分函数 $v_s: \mathbb{P}_{\mathbb{D}} \rightarrow \{\text{true}, \text{false}\}$ 。

假设 1 对于任何领域 $\mathbb{D}$ 的概念化 $\mathbb{C}$ ，都存在恰好一个 BPC $\overline{\mathbb{C}}$ ，它描述了内部化 $\mathbb{C}$ 的代理评估关于 $\mathbb{D}$ 的命题的能力。

基于这一假设，如果断言命题在任何属于概念化的状态中为真，则我们可以将本体版本定义为一个给定概念化的正确规范。然而，正如我们在第 3 节中讨论的，本体的发展通常涉及具有不同概念化的一组人。为此，我们按如下方式推广这一概念：如果本体版本与每个单个概念化一致，并且本体断言的每个命题至少根据一个概念化为真（参见定义 8），则本体版本是概念化集的正确规范。本体版本可能包含错误。因此，并非本体版本的每个规范都是正确的。尽管如此，本体版本仍然是概念化的规范（参见定义 9）。

定义 8（概念化正确规范）设 $\mathbb{D}$ 为一个领域， $\mathbb{C}$ 为 $\mathbb{D}$ 的一个概念化，使得 $\overline{\mathbb{C}} = \langle \mathbb{S}, v \rangle$ ， $\mathfrak{C} = \{\mathbb{C}_1, \dots, \mathbb{C}_n\}$ 为 $\mathbb{D}$ 的一组概念化，使得对于任何 $\mathbb{C}_i \in \mathfrak{C}$ ，其对应的 BPC $\overline{\mathbb{C}}_i = \langle \mathbb{S}_i, v_i \rangle$ ，以及本体版本 $\mathbf{o}$ 关于 $\mathbb{D}$ 。

$\mathbf{o}$ 与 $\mathbb{C}$ 一致当且仅当 $\mathbf{o}$ 是一个关于 $\mathbb{D}$ 的本体版本，并且对于任何事件状态 $s \in \mathbb{S}$: $v_s(p) \neq \text{为假}$ ，对于任何 $p \in \mathbb{P}^{\mathbf{o}}$ 。

$\mathbf{o}$ 是 $\mathfrak{C}$ 的一个正确规范当且仅当 $\mathbf{o}$ 是一个关于 $\mathbb{D}$ 的本体版本并且

1. $\mathbf{o}$ 与 $\mathbb{C}_i$ 一致，对于任何 $\mathbb{C}_i \in \mathfrak{C}$ ；
2. 对于任何 $p \in \mathbb{P}^{\mathbf{o}}$ ，存在一个 $\mathbb{C}_i \in \mathfrak{C}$ ，使得对于所有 $s \in \mathbb{S}_i$ ， $v_s(p) = \text{为真}$ 。

定义 9（概念化规范）本体版本 $\mathbf{o}$ 指定了概念化 $\mathfrak{C}$ 当且仅当 $\mathbf{o}$ 的作者意图让 $\mathbf{o}$ 成为 $\mathfrak{C}$ 的一个正确规范。

定义 8 和 9 解释了为什么知识分工不会阻碍本体发展，因为人们可以共同开发本体，即使他们对该领域的概念化不同。不同的本体开发者甚至可能拥有相互矛盾的对该领域的概念化，只要本体对有争议的命题保持中立。

states of affair in the domain that the conceptualization is about and provides for each state of affair a partial mapping from propositions to *true* or *false*. The mapping is partial, since we allow incomplete conceptualizations of a domain.

Definition 7 (Behavioral Profile of a Conceptualization (BPC)) A BPC of a domain $\mathbb{D}$ is an ordered pair $\langle \mathbb{S}, v \rangle$ where

1. the set of states of affair $\mathbb{S}$ is non-empty;
2. the valuation function v assigns a partial function $v_s: \mathbb{P}_{\mathbb{D}} \rightarrow \{\text{true}, \text{false}\}$, to each state of affair $s \in \mathbb{S}$.

Assumption 1 For any conceptualization $\mathbb{C}$ of a domain $\mathbb{D}$ there exists exactly one BPC $\overline{\mathbb{C}}$ of $\mathbb{D}$ that characterizes the ability of an agent that internalizes $\mathbb{C}$ to evaluate propositions about $\mathbb{D}$.

Based on this assumption, we could define a ontology version to be a correct specification of a given conceptualization if the asserted propositions are true under any state of affair that is part of the conceptualization. However, as we have discussed in Section 3, the development of ontologies typically involves a number of people with varying conceptualizations. For this reason, we generalize the notion as follows: an ontology version is a correct specification of a set of conceptualizations, if it is consistent with each individual conceptualization and every proposition that is asserted by the ontology is true according to at least one conceptualization (see Def. 8). An ontology version may contain bugs. Thus, not every specification by an ontology version is correct. Nevertheless, the ontology version is still a specification of the conceptualizations (see Def. 9).

Definition 8 (Correct Specification of Conceptualizations) Let $\mathbb{D}$ be a domain, $\mathbb{C}$ a conceptualization of $\mathbb{D}$ such that $\overline{\mathbb{C}} = \langle \mathbb{S}, v \rangle$, $\mathfrak{C} = \{\mathbb{C}_1, \dots, \mathbb{C}_n\}$ a set of conceptualizations of $\mathbb{D}$, such that for any $\mathbb{C}_i \in \mathfrak{C}$ the corresponding BPC $\overline{\mathbb{C}}_i = \langle \mathbb{S}_i, v_i \rangle$, and $\mathbf{o}$ an ontology version about $\mathbb{D}$.

$\mathbf{o}$ is consistent with $\mathbb{C}$ iff $\mathbf{o}$ is an ontology version about $\mathbb{D}$ and for any state of affair $s \in \mathbb{S}$: $v_s(p) \neq \text{false}$, for any $p \in \mathbb{P}^{\mathbf{o}}$.

$\mathbf{o}$ is a correct specification of $\mathfrak{C}$ iff $\mathbf{o}$ is an ontology version about $\mathbb{D}$ and

1. $\mathbf{o}$ is consistent with $\mathbb{C}_i$, for any $\mathbb{C}_i \in \mathfrak{C}$;
2. for any $p \in \mathbb{P}^{\mathbf{o}}$, there is a $\mathbb{C}_i \in \mathfrak{C}$, such that for all $s \in \mathbb{S}_i$, $v_s(p) = \text{true}$.

Definition 9 (Specification of Conceptualizations) An ontology version $\mathbf{o}$ specifies the conceptualizations $\mathfrak{C}$ iff the authors of $\mathbf{o}$ intend $\mathbf{o}$ to be a correct specification of $\mathfrak{C}$.

Definitions 8 and 9 explain why a division of intellectual labor does not prevent ontology development, because it is possible for people to develop an ontology together, even if they do not share the same conceptualization of the domain. The different ontology developers may even possess conceptualizations of the domain that contradict each other, as long as the ontology is non-committal concerning the contested propositions.

4.5 概念主义与实在论

虽然概念主义者会提出本体是否准确代表我们对该领域概念化的疑问，但现实主义者主要关注本体是否准确代表现实。这两个问题都相当合理且相关。我们称对真实领域的概念化为正确的，如果在概念化所考虑的所有情况下，该情况中的赋值函数与现实不冲突（参见定义10）。可以证明，正确概念化的正确规范，是该领域的正确本体（定理1）。

定义 10（正确的概念化）设 $\mathbb{D}$ 是一个真实领域， $\mathbb{C}$ 是 $\mathbb{D}$ 的一个概念化，且 $\overline{\mathbb{C}} = \langle \mathbb{S}, v \rangle$ 。 $\mathbb{C}$ 是 $\mathbb{D}$ 的一个正确概念化当且仅当 (1) $\mathbb{S} \subseteq \mathbb{D}$ 且 (2) 对于任何事态 $s \in \mathbb{S}$ 和任何命题 $p \in \mathbb{P}_{\mathbb{D}}$ ：
 (a) 如果 $v_s(p) = \text{为真}$ ，则 $p \in s$ ；
 (b) 如果 $v_s(p) = \text{为假}$ ，则 $p \notin s$ 。

定理 1 如果 $\mathfrak{C}$ 是真实领域 $\mathbb{D}$ 的一组正确概念化，且 $\mathfrak{o}$ 是 $\mathfrak{C}$ 的一个正确规范，则 $\mathfrak{o}$ 是 $\mathbb{D}$ 的一个正确本体版本。

5 相关工作

在信息系统领域，“本体”一词最早由 Mealy [17] 于 1967 年使用。⁶ 然而，更具意义的是 1969 年发表的《来自人工智能立场的某些哲学问题》[16]，其中 McCarthy 和 Hayes 认为通用智能机器需要具有形而上学和认识论上充分的关于世界的表示，并提出了情境算子作为表示语言。尽管该术语在 [16] 今天我们称之为“本体”，但 [16] 中提出的愿景导致了 Hayes 对朴素物理学的相关工作，特别是他 1978 年发表的论文《朴素物理学 I: 液体本体》[12]，这似乎是第一篇标题中包含“本体”的计算机科学论文。[12] 中没有包含“本体”的明确定义，但第一句话说明该文包含 “[...] 常识知识的形式化 [...]”。

因此，“本体”这一术语在计算机科学中早已确立，而 1992 年 Tom Gruber 提出了定义：“本体是对概念化的具体规范”[4]。不久之后，Gruber 补充了规范必须是明确的要求 [5, 6]。Borst [3] 补充了规范必须是形式化的，且概念化必须是共享的条件。Studer [24] 将这些结合，将本体定义为对共享概念化的形式化、明确的规范。

⁶ 为了简洁起见，我们限制了“本体”一词历史的讨论，特别是省略了其在哲学中的历史。更详细的说明可以在 [22] 中找到。

4.5 Conceptualism vs. Realism

While the conceptualist raises the question whether an ontology accurately represents our conceptualizations of a given domain, the realist is mainly concerned with the question whether an ontology represents reality accurately. Both questions are quite reasonable and related. We call a conceptualization of a real domain correct, if in all situations that are considered by the conceptualization, the valuation function in that situation does not conflict with reality (see Def. 10). One can prove that a correct specification of a correct conceptualization, is a correct ontology of the domain (Theorem 1).

Definition 10 (Correct conceptualization) Let $\mathbb{D}$ be a real domain, $\mathbb{C}$ be a conceptualization of $\mathbb{D}$, and let $\overline{\mathbb{C}} = \langle \mathbb{S}, v \rangle$. $\mathbb{C}$ is a correct conceptualization of $\mathbb{D}$ iff (1) $\mathbb{S} \subseteq \mathbb{D}$ and (2) for any state of affair $s \in \mathbb{S}$ and any proposition $p \in \mathbb{P}_{\mathbb{D}}$:
 (a) if $v_s(p) = \text{true}$, then $p \in s$;
 (b) if $v_s(p) = \text{false}$, then $p \notin s$.

Theorem 1 If $\mathfrak{C}$ is a set of correct conceptualizations of the real domain $\mathbb{D}$ and $\mathfrak{o}$ is a correct specification of $\mathfrak{C}$, then $\mathfrak{o}$ is a correct ontology version of $\mathbb{D}$.

5 Related Work

In the context of information systems “ontology” was first used in 1967 by Mealy [17].⁶ However, arguably more significant is *Some Philosophical Problems from the Standpoint of Artificial Intelligence* [16] from 1969, in which McCarthy and Hayes argue that general intelligent machines require *metaphysically and epistemologically adequate representations* of the world, and propose the situation calculus as a representation language. Although the term is not used in [16], today we would call these representations ‘ontologies’. The vision presented in [16] lead to Hayes’ work on naive physics, in particular his 1978 paper “Naive Physics I: Ontology for Liquids” [12], which seems to be the first computer science paper with “ontology” in the title. [12] does not contain an explicit definition of “ontology”, but the first sentence states that the essay contains “[...] a formalisation of common-sense knowledge [...]”.

Thus, the term “ontology” was already established in computer science, when in 1992 Tom Gruber proposed the definition “An ontology is a specification of a conceptualization” [4]. Shortly afterward Gruber added the condition that the specification is explicit [5, 6], Borst [3] added the conditions that the specification needs to be formal and the conceptualization needs to be shared. Studer [24] combined these and defines an ontology as a *formal, explicit specification of a shared conceptualization*.

⁶ For the sake of brevity, we limit our discussion of the history of the term “ontology” and, in particular, omit its history in philosophy. A more detailed account can be found in [22].

Gruber初始定义的所有修改都值得怀疑。在 [5]中, Gruber很清楚地从形式化软件规范社区借用了“规范”一词, 该社区使用“Z”等形式化方法来描述软件系统。在 [5] 中, KIF被用作规范语言。因此, Gruber意义上的任何规范都是形式化的。此外, 由于不清楚隐式KIF文档会是什么样子, 显式条件似乎也是多余的⁷。Borst主张包含共享概念化条件, 因为 “[...] 当本体所规范的概念化不被普遍接受时, 重用本体的能力将几乎为零。” [3]。该论证似乎基于一个基本假设, 即共享概念化是共享使用或重用的前提条件。正如我们在第3.3节和第4.4节中所论证的, 本体可以表示来自那些对领域概念化不同的知识持有者的知识。

Guarino 认为“规范”一词过于强烈, 建议将定义弱化为“给出概念化显式、部分描述的逻辑理论” [10]。Smith 和 Welty 认为 Gruber 的定义过于宽泛, 因为它涵盖了从目录、术语表到一组逻辑约束的多样化工件。 [23]

对我们对基于概念化的本体定义的批评的一个显著例外是 Guarino 等人在 [11], 中进行的开创性工作, 该工作基于 [10, 8] 构建。它提出了一种基于概念化形式理论的“本体”定义。简而言之, [11] 将本体识别为一组公式, 其模型近似于概念化, 而概念化则被识别为一种意向性的一阶关系结构。由于 [11] 将本体识别为一组公式, 它不能解释本体变化的能力, 并且未能解决词汇选择和注释的作用。此外, 它还假设本体基于共享的概念化, 这未能解释本体开发过程中的分工。另外两个问题在于, 它没有解释“近似”一词的预期含义, 以及将概念化与意向性的一阶结构相等同, 这导致了若干技术和哲学问题。[11] 的详细批评可以在 [18] 中找到。

6 结论

在本文中, 我们提出了“本体”的一个替代性定义。该定义基于一个形式化理论, 涉及两个关键概念: 一个带注释的逻辑理论和一个预期解释。该定义不对本体的逻辑语言或所用注释类型做任何假设。此外, 它不仅与概念主义和实在论兼容, 还允许证明正确的概念化规范是一个正确的本体(版本)。该定义的丰产性进一步

⁷遗憾的是, 在 [5, 6] Gruber 并没有讨论他为什么添加了显式条件。

All of the modifications of Gruber’s initial definition are dubious. In [5] it is quite clear that Gruber borrows the term “specification” from the formal software specification community, which uses formalisms like “Z” to describe software systems. In [5] KIF is used as specification language. Thus, any specification in the sense of Gruber is formal. Further, since it is unclear what an implicit KIF document would look like, the explicit condition seems redundant as well.⁷ Borst argues for including the shared-conceptualization-condition, because “[...] the ability to reuse an ontology will be almost nil when the conceptualization it specifies is not generally accepted.” [3]. The argument seems to be based on the underlying assumption that a shared conceptualization is a precondition for shared use or reuse. As we have argued in Sections 3.3 and 4.4, an ontology may represent knowledge from people whose conceptualization of the domain differs.

Guarino argues that the term “specification” is too strong and suggests to weaken the definition to “logical theory which gives an explicit, partial account of a conceptualization” [10]. Smith and Welty argue that Gruber’s definition is too wide, since it encompasses a diverse group of artifacts ranging from catalogs, over glossaries to a set of logical constraints. [23]

One notable exception to our criticism of conceptualization-based definitions of ontology is the seminal work by Guarino et al. in [11], which builds on [10, 8]. It presents a definition of “ontology” based on a formal theory of conceptualizations. In short, [11] identifies an ontology with a set of formulas whose models approximate a conceptualization, which is identified with an intensional first-order relational structure. Since [11] identifies ontologies with sets of formulas, it does not explain the ability of ontologies to change, and fails to address the role of the choice of the vocabulary and the annotations. Further, it also assumes that ontologies are based on shared conceptualizations, which fails to account for the division of labor during ontology development. Two other problematic aspects are the lack of explanation of what “approximates” is supposed to mean, and the identification of conceptualizations with intensional first-order structures that lead to several technical and philosophical issues. A detailed critique of [11] may be found in [18].

6 Conclusion

In this paper we presented an alternative definition of “ontology”. It is based on a formal theory that involves two key notions: an *annotated logical theory* and an *intended interpretation*. The definition does not make any assumptions about the logical language of the ontology or the type of annotations that are used. Further, it is not only compatible with both conceptualism and realism, but allows showing that a correct specification of correct conceptualizations is a correct ontology (version). The fruitfulness of the definition is further

⁷Unfortunately, in [5, 6] Gruber does not discuss why he added the explicit condition.

通过解释词汇和注释选择在建立预期解释中的作用，以及说明如何在没有共同概念化的情况下实现合作本体开发，得到了进一步说明。

未来，我们计划研究该方法对本体论评估的影响（公理的完备性、公理化不足、缺乏文档）。此外，我们计划将其应用于特定逻辑（例如，FOL），这将使我们能够将逻辑的形式语义与其预期解释联系起来。

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illustrated by an explanation of the role of the choice of the vocabulary and annotations for establishing an intended interpretation, and an account of how cooperative ontology development is possible without a shared conceptualization.

In the future we are planning to study the approach’s consequences for ontology evaluation (completeness of axioms, underaxiomatization, lack of documentation). Further, we are planning to instantiate it for particular logics (e.g., FOL), which will enable us to relate the formal semantics of the logic to the intended interpretations.

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